

Haris Spyridis

Computational Biologist · Bioinformatician

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PROFILE

Computational genomics researcher with experience in whole-genome sequencing (WGS), RNA-seq analysis, variant discovery, comparative genomics and reproducible pipeline development (Nextflow, Git and HPC). I combine a strong wet-lab foundation in molecular biology with growing expertise in deep learning for biological sequence data. Currently interested in computational methods for large-scale genomic and metagenomic data, with an emphasis on evolutionary and comparative questions.

EXPERIENCE

Open-Source Contributor - nf-core

Mar 2026 - present

- Contributing to the development, testing and documentation of the **taxprofiler** pipeline for metagenomic taxonomic profiling.
- Admitted to the nf-core/Nextflow ambassador program (July 2026).

Co-authored Research Proposal

With Prof. Marek Mutwil, University of Copenhagen

Aug 2025

- Designed a proposal to benchmark CNN and transformer-based sequence embeddings for cross-species prediction of expression and metabolic responses in plants.

Operations Support - Family business, Athens

Jan 2024 - Sep 2025

- Supported operations in a family-run engineering company while pursuing opportunities in bioinformatics and continuing self-directed training in computational genomics and ML.

Bioinformatics research intern

Mar 2023 - Aug 2023

Wageningen University, Biosystematics group - Collaboration with John Innes Centre

- Adapted a comparative genomic [pipeline](#) to detect introgressions across lettuce and wild relatives.
- Performed phylogenomic analyses, gene annotation, and GO/pathway enrichment to interpret candidate introgression regions.
- Contributed to reproducible and scalable workflows using Snakemake, Git and HPC.

Laboratory Intern

Benaki Phytopathological Institute, Athens

Jun 2017 - Sep 2017

- Performed molecular assays to study genetic diversity of plants species and evaluated infested plants.

EDUCATION

MSc (Plant) Biotechnology - Functional Genomics

Wageningen University, Netherlands

Sep 2021 - Aug 2023

Started a 2nd MSC focused on bioinformatics to build computational foundations in genomics and ML

MSc Thesis Phytopathology group; collaboration with the Department of Genetics:

“Genetic loci underlying sensitivity to the phytotoxin botrydial”.

- Performed WGS variant discovery coupled with bulk segregant analysis to identify candidate loci, underlying a polygenic trait architecture.

- Integrated KASP genotyping with quantitative phenotyping to assess genotype-phenotype associations.

MSc Plant Science - Plant Pathology & Entomology

Wageningen University, Netherlands

Mar 2020 - Sep 2021

MSc Thesis ([MICROP](#) consortium, 5 Dutch Universities): Plant - insect - microbe interactions: Soil legacy effects on plant resistance to insects.”

BSc in Plant Science (Integrated MSc, 300 ECTS)

Aristotle University of Thessaloniki, Greece

Sep 2013 - Jul 2018

PUBLICATIONS

- Stamouli S. et al. “nf-core/**taxprofiler** and nf-core/createtaxdb provide scalable and reproducible metagenomic multi-classifier database construction and taxonomic profiling.” Submitted to *Microbial Genomics* (May 2026). **Spyridis, H.** (co-author)
- Aragon M. ,**Spyridis H.**, et al. (2026) “Jasmonic and Salicylic Acid Pathways Shape the Rhizosphere Microbiome, affecting Aphid Herbivory and Soil-Mediated Insect-Pitlant Interactions”.
Plant & cell Physiology. DOI: [10.1093/pcp/pcag048](https://doi.org/10.1093/pcp/pcag048)

SKILLS

Computational Genomics	WGS processing Variant discovery Comparative & phylogenomic analysis RNA-seq differential expression (DESeq2) GO/pathway enrichment analysis (Bioconductor) metagenomic taxonomic profiling
Programming	Python R Bash scripting
Workflows	Nextflow Snakemake Git HPC/Linux environments
Machine/Deep Learning	TensorFlow PyTorch Neural networks for biological sequence data familiarity with CNNs and Transformer-based models
Molecular techniques	DNA/RNA extraction (q)PCR, RT-qPCR Cloning techniques ROS assays CRISPR/Cas9 Gel electrophoresis ELISA KASP assays
Languages	Greek (native) English (fluent) Dutch (basic)

CONTINUING EDUCATION

- [Genomic Data Science Specialization](#) - Johns Hopkins University
- [Machine Learning Specialization](#) - Stanford University
- [Deep learning Specialization](#) - DeepLearning.AI

AWARDS

- Academy Consultancy Training (ACT) Project, WUR - Developed a decision support tool for plant protection using basic substances; ranked top 6 out of 211 projects (2021)

ADDITIONAL

Student Assistant - DeepLearning.AI

Jan 2026 - present

- Reviewed programming assignments and notebooks for ML/DL courses; identified and reported bugs in course materials.
- Support learners with debugging and conceptual questions in Python, Tensorflow and ML topics.

Mandatory military service, Greece

Sep 2018 - Sep 2019